AWS VPC PEERING PROJECT REPORT

EWB Courses — AWS

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Project focus	Two VPCs, VPC peering, routing, security groups and EC2 instances

Submission note

This report documents the AWS configuration recorded during project work. The VPC peering connection was reported as Active. End-to-end connectivity between App-A and App-B has not been verified. Resource screenshots will be added separately by the student.

Prepared for project documentation and GitHub portfolio use

1. Abstract

This project documents two Amazon Virtual Private Clouds (VPCs) and a VPC peering connection intended to enable private routing between workloads in separate VPCs. The recorded environment includes subnets, route tables, an Internet Gateway, security groups, a bastion EC2 instance and two application EC2 instances. The peering connection was reported as Active. App-A Session Manager remains unavailable and App-A/App-B connectivity has not been verified; these limitations are stated explicitly.

2. Objectives

• Create two VPCs with non-overlapping IPv4 CIDR ranges.

• Create subnets and launch EC2 instances for bastion and application roles.

• Establish VPC peering between the two VPCs.

• Configure routes for traffic to the remote VPC through the peering connection.

• Configure security groups for intended SSH and ICMP traffic.

• Document the setup and identify remaining validation tasks.

3. AWS environment and resources

AWS Region: **ap-southeast-2 (Sydney)**. Values below are recorded from the project conversation; verify them against your screenshots before submission.

Resource	Recorded name / value	Purpose / notes
VPC A	ewb-vpc-a — 10.10.0.0/16	Contains bastion and App-A
VPC B	ewb-vpc-b — 10.20.0.0/16	Contains App-B
Peering connection	ewb-a-b-peer	Reported status: Active
Internet Gateway	ewb-igw-a	Attached to VPC A
Public subnet A	ewb-subnet-a-public — 10.10.1.0/24	Subnet ID recorded: subnet-0e82b80d75937de4c
Bastion	ewb-bastion — 10.10.1.15	Public IP recorded: 3.106.117.86
App-A	ewb-app-a — 10.10.2.40	Private application instance
App-B	ewb-app-b — 10.20.1.168	Private application instance

For a public repository, consider redacting the bastion public IP and subnet ID if they are not required by the assignment.

4. Architecture overview

Intended architecture: a bastion host in VPC A's public subnet, App-A in a private subnet in VPC A, and App-B in a private subnet in VPC B. The two VPCs are connected through the peering connection. This is a textual architecture summary, not a live AWS-generated diagram.

VPC A — 10.10.0.0/16	VPC Peering	VPC B — 10.20.0.0/16
Public subnet 10.10.1.0/24 Bastion 10.10.1.15 Internet Gateway ewb-igw-a	ewb-a-b-peer Reported Active	Private subnet 10.20.1.0/24 App-B 10.20.1.168
Private subnet 10.10.2.0/24 App-A 10.10.2.40		

Connectivity status: App-A/App-B communication has not been verified. This diagram describes intended architecture, not a successful ping or SSH result.

5. Routing configuration

Route	Recorded target	Purpose
10.10.0.0/16	local	Traffic inside VPC A
10.20.0.0/16 (from VPC A)	VPC peering connection	Route to VPC B
0.0.0.0/0 (public subnet)	ewb-igw-a	Internet route for public subnet
10.20.0.0/16	local	Traffic inside VPC B
10.10.0.0/16 (from VPC B)	VPC peering connection	Route to VPC A

Attach actual route-table screenshots and correct this section if the final console values differ.

6. Security groups

Security group	Recorded inbound rules / intent
ewb-sg-bastion	SSH TCP 22 restricted to the user's public IP /32 at the time of setup; this address can change.
ewb-sg-app-a	SSH TCP 22 from bastion private IP 10.10.1.15/32; ICMP rule recorded for App-B 10.20.1.168/32.
ewb-sg-app-b	SSH TCP 22 from bastion private IP 10.10.1.15/32; ICMP rule recorded for App-A 10.10.2.40/32.

7. Implementation and verification status

Item	Status
VPCs and EC2 instances	Created according to recorded setup
VPC peering connection	Reported Active in AWS Console
Route tables and security groups	Configured according to recorded setup; screenshots should be attached
Bastion Session Manager	Previously opened successfully; hostname output was obtained
App-A Session Manager	Not connected; ewb-ssm-role attached but instance still did not register/connect
App-A ↔ App-B connectivity	Not tested / not verified
Successful ping or SSH between app instances	No successful result recorded

8. Limitations and next steps

• Troubleshoot App-A Systems Manager access if terminal access is required.

• When access is available, test connectivity between App-A (10.10.2.40) and App-B (10.20.1.168).

• Save genuine terminal output if the connectivity test is later completed.

• Check the assignment PDF to determine whether successful connectivity evidence is mandatory.

• Review AWS billing and stop or delete billable resources when they are no longer needed, subject to assignment requirements.

9. Evidence and screenshots

Add the screenshots you have successfully captured to the repository's screenshots folder. The final two evidence items from the previous checklist have been removed from this index because they were not available/working for your submission.

Evidence	Recommended filename	Status
VPC A details	01-vpc-a.png	Add screenshot
VPC B details	02-vpc-b.png	Add screenshot
Peering connection showing Active	03-vpc-peering.png	Add screenshot
Subnets list	04-subnets.png	Add screenshot
VPC A public route table	05-route-table-a.png	Add screenshot
VPC B private route table	06-route-table-b.png	Add screenshot
Security group inbound rules	07-security-groups.png	Add screenshot
EC2 instances and states	08-ec2-instances.png	Add screenshot

10. Conclusion

The project records two AWS VPCs with separate CIDR ranges, a reported Active peering connection, route-table entries, security groups, and three EC2 instances. The bastion Session Manager terminal was accessible. App-A Session Manager remains unavailable and communication between App-A and App-B has not been verified. The infrastructure setup is documented, but end-to-end connectivity validation remains outstanding.

11. Public GitHub safety checklist

■ Do not upload .pem private keys, AWS access keys, credentials, tokens or passwords.

- Review screenshots for account identifiers, personal public IPs and other sensitive information.
- Share the course PDF only if its terms allow redistribution.
- Keep connectivity marked as unverified until there is genuine test output.
- Review any running AWS resources to avoid unexpected charges.

Report integrity note

This report is based on details recorded during the conversation and is not an independent AWS audit. Confirm all values against the AWS Console and attach your genuine screenshots before submission. No successful connectivity test is claimed.